Finding a Cause

A. Overconsumption

Overconsumption refers to the excessive use of resources, goods, and services, leading to detrimental environmental, social, and psychological effects

SDG's

12- Responsible Consumption and Production
13- Climate Action

Issue #1- CONSUMER CULTURE

Issues of Interest	1. **Overconsumption Linked to Consumer Culture** The excessive purchasing of goods and services due to the natural influences of the consumeristic economy we live in. Another major influence is socio-economic status, where buying new things and keeping up with trends signals status and hierarchy.
Significance to ME	1. **DIRECT Experience** Consumer culture and the need to constantly buy things for self-gratification are experiences that I have felt. I also feel like there are no apps/websites that target this issue despite it only GROWING as technology becomes increasingly accessible and advanced. 2. **Direct Climate Issues** The exacerbation of overconsumption doesn't lead to just temporary issues for the buyer, it also contributes to unhealthy waste that negatively affects our climate. 3. **Stopping the Trend** It seems like apps struggle to truly grab the root of the issue, which is the widespread influence of social media. Maybe limiting apps are not enough, but instead encouraging critical thinking and better choices when thinking about buying something (encouraging better financial practices)
Competitors	1. **Traditional ShopFreeze apps** Ex. I Am Sober, PocketGuard 2. **Apps that MANAGE spending** Ex. YNAB, GoodBudget 3. **Pre-Purchase Interventions** Ex. Envelope, SelfControl
Informational Resources	**CONSUMER CULTURE LINK** **REDDIT THREAD: Breaking the Cycle of Overconsumption** **REDDIT THREAD: Women Being the Biggest Catalyst for Overconsumption** **WOMEN CONSUMER FACTS**

Issue #2- OVERCONSUMPTION IN STEM

Issues of Interest	1. **Overconsumption in STEM** The excessive purchase and waste of electronic and raw materials due to technological updates, lack of necessity, or quality of the product.
Significance to ME	1. **"One man's trash is another man's treasure"** Dumb as it sounds, a lot of the "useless materials" and "outdated electronics" can actually be useful to other STEM programs and even students trying to make independent projects themselves. Having an app could help take unwanted materials from established programs and schools to other programs and independent students. 2. **Integration w/ popular STEM clubs** A lot of the same electronics/materials trend across clubs. Like Roboboat, FRC, AVR, etc. will use the same types of electronic and raw materials. If teams could help each other out and work together to be more sustainable where material overflow is common, that could be cool. 3. **Prevents Overconsumption while PROMOTING Accessibility** It doesn't just stop at overconsumption. Teams that are just starting out and aren't backed by a lot of funding can build a network with more experienced, established teams while also getting resources for free/cheap, fast.
Competitors	1. **School-designated Marketplace** Ex. The Buy Nothing Project 2. **STEM Surplus Sites** Ex. LEDP, BioLink Depot 3. **Social Media Sites** Ex. Instagram, Facebook Marketplace, Discord, Chief Delphi
Informational Resources	**MIT: Tech Waste in 2021** • "I don't know what I would do if someone gave me a pile of old electronics pieces, but they've created all these amazing machines, even 3D printers, from recycled tech" • "Lack of awareness is probably the biggest issue we have in regard to the e-waste problem. If we're aware it's a problem, solutions can start flowing in" **RESEARCH: Does STEM = demise of Env. Sustainability**

App Ideation

(Each bullet point is a different app, not just a different feature)

Issue #1- CONSUMER CULTURE	**Issue #2- OVERCONSUMPTION IN STEM**
• Tracks spending habits, bans shopping sites after certain usage time, limits/tracks expenditures and spits out data • Encourages user to find hobbies, limits shopping apps/sites after certain usage time, notifies/keep tracks with users hobbies • Upscale app, selling old clothes/used items • Busting trends by having actual feedback and assessments on current fads and trends, a social site that discusses the environmental and social harm of widespread trends	• Designated app for popular clubs and teams to collaborate and share resources • App that can be used by school districts to support independent school projects by giving away unwanted electronics/materials • Upload inventory of workshop and flag unwanted items, where app will notify/find places that need the resource you don't

AI Overview (TOP PICKS)

- **#7** (workshop inventory → auto-match unwanted items to places that need them) is the strongest overall. It's not just a static listing board — a matching/notification algorithm gives you real technical complexity to show off (which Congressional App Challenge judges specifically reward — they like seeing actual logic, not just CRUD screens), and it pairs with a strong personal-story angle ("Significance to ME" / direct experience in STEM clubs) that Technovation's judging rubric leans on heavily.
- **#3** (resale app for old clothes/used items) is the safest, most "guaranteed to work" option — simple, well-understood marketplace mechanics, easy to demo cleanly in a CAC video. Its risk is differentiation (Poshmark/Depop already exist), so for Technovation you'd want to narrow it — e.g., school-only or local-community resale — to make the impact story land.
- **#6** (district-level electronics redistribution) is a good middle ground: broader social impact than #7 (helps whole districts, not just clubs), still a feasible matching/marketplace build, and ties cleanly to SDG 12.

BOTTOM LINE: I'd build around #7 or a #7+#6 hybrid as the primary pitch (best complexity-to-impact ratio for both competitions), with #3 as the fallback if the matching algorithm proves too ambitious for the timeline.

B. Educational Efficiency and Efficacy

Optimizing the consumption of knowledge and the improvement of a student's mental state while learning

SDG's

4- Quality Education

Issue #1- LEARNING ABOUT LEARNING

Issues of Interest	1. **Learning about Learning** Students often have a difficult time retaining information for a long time and being academically efficient because they do not know how to absorb knowledge (or study). With the prevalence of short-term content and addictions to technological apps that provide gratification, studying can seem pointless and boring without a sure fire plan.
Significance to ME	1. **Studying is Fake** So many academically successful students don't know how to efficiently study until they reach COLLEGE. The younger a child is exposed to be efficient, the more it nurtures a love for learning and a passion to continue more productive bouts of knowledge. 2. **Improving Self Efficacy** A lot of students experience mental fatigue and stress because of studying. Learning how to exercise efficient strategies and optimize time will help lessen the stress and bring better results. 3. **Encourages Active Learning** Improving a student's metacognition helps children take more active roles in their learning. From being more vocal in educational settings to doing outside research to things they love, attaining knowledge and grasping a concept will become less burdensome and more exciting.
Competitors	1. **Meta-cog Apps** Ex. Ankii, Kinuu, Knowt 2. **Search Engines/LLM's** Ex. Google, Claude, Chat GPT 3. **App Freezers/Limiters** Ex. Freedom, Opal, AppBlock
Informational Resources	My ANNOTATED BIB for this app….

App Ideation

(Each bullet point is a different app, not just a different feature)

Issue #1- LEARNING ABOUT LEARNING
• Sets students up on a learning schedule with learning strategies they think will work, when detects a certain amount of usage on limit, banned, or inhibited apps, will send a notification that apps will shut down soon and they should start studying. Focus modes will also be implemented, users can choose specific modes that will target and limit apps recommended by the app or b/c they link with distractions (social media, messaging apps etc.). After studying, users will do a quick survey, and based on that their learning plan will be adjusted

AI Overview (TOP PICKS)

What it actually requires, broken into pieces:

1. **Usage detection** — knowing how much time is spent in distracting apps
2. **App blocking/throttling** — shutting down or restricting those apps
3. **Scheduling engine** — building a study plan around learning strategies
4. **Focus modes** — preset bundles of restrictions tied to specific distraction categories
5. **Adaptive feedback loop** — post-study survey that adjusts the plan going forward

Feasibility reality check: Pieces 1 and 2 are the risky part. True OS-level app blocking on iOS requires Apple's Screen Time API/Family Controls entitlement, which is genuinely hard for a student team to get approved for — that's the same wall that sank idea #2 from the Overconsumption section. Android is a bit more open (Usage Stats + Accessibility Service can detect/limit app usage without special entitlements), but it's still nontrivial.

The fix that makes this very buildable: pivot the "blocking" mechanic to the browser instead of the OS. A Chrome extension that detects time-on-site and blocks/redirects distracting sites during a focus session is something a student can build in a weekend, no entitlements, no app store review. You lose native mobile polish, but you gain a fully working demo — which matters more for both CAC and Technovation than feature breadth.

- The adaptive survey → adjusted plan loop is a legitimate "smart" feature judges respond well to, and it doesn't require real ML — a rule-based recommendation engine ("3 sessions of low focus on mode X → suggest mode Y") sells just as well in a pitch and is far more buildable on a deadline.
- SDG 4 (Quality Education) is explicit and the personal story ("studying is fake until college") is the kind of direct-experience narrative Technovation's rubric rewards.
- It's a universal problem (every student struggles with this) rather than a niche one, which is an easier "market" pitch than the STEM-club resource apps.